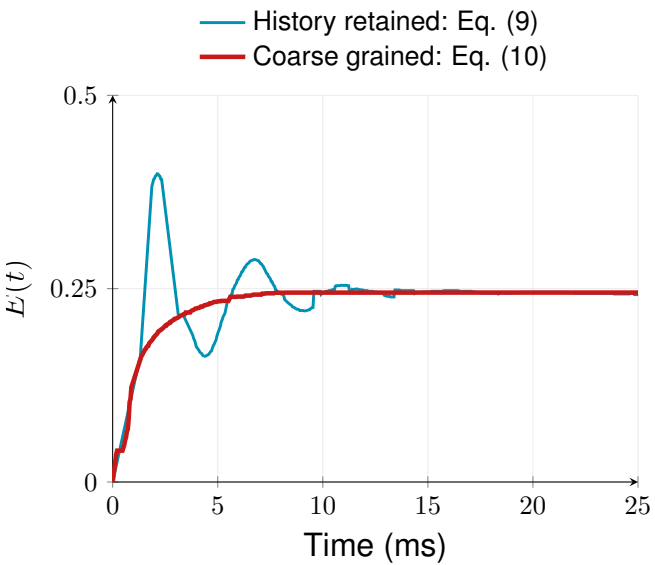
Population activity: temporal coarse graining

Excitatory population only: explicit memory versus temporal averaging



Digitized redrawing of Fig. 3; $r = 3\,\mathrm{ms}$.

History retained — Equation (9)

$$\tau \dot{E}(t) = -E(t) + \left[1 - \int_{t-r}^{t} E(s)\,ds\right] \times S_e\left(\int_{-\infty}^{t} e^{-\alpha(t-s)}[c_1 E(s) + P(s)]\,ds\right).$$

Coarse grained — Equation (10)

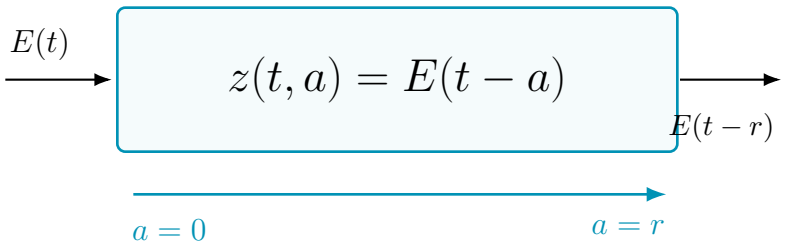
$$\tau \dot{\bar{E}}(t) = -\bar{E}(t) + (1 - r\bar{E}(t)) \times S_e\big(k[c_1 \bar{E}(t) + P(t)]\big).$$

Wilson and Cowan (1972), Fig. 3 and Eqs. (9)-(10). Curves digitized from the published scan.

Retaining population memory with transport

An equivalent realization of Equation (9), with compatible initial history

Store the recent activity history



Memory age a, not tissue position.

Refractory memory retained.
Synaptic filtering retained.
No temporal coarse graining.

Transport PDE and refractory fraction

$$\partial_t z + \partial_a z = 0, \qquad a \in [0, r],$$

$$z(t,0) = E(t),$$

$$R(t) = \int_0^r z(t,a)\, da.$$

Coupled ODEs

$$\dot{U} = -\alpha U + c_1 E + P,$$

$$\tau \dot{E} = -E + (1 - R) S_e(U).$$

Transport formulation derived from Wilson and Cowan (1972), Eq. (9). The extra coordinate is memory age.